THE CUP SQUARE $\kappa : K(\mathbb{Z},2) \to_* K(\mathbb{Z},4)$ BY DELOOPING

Purpose and status. Informal construction for step §2b of `doc/pi3-s2-postnikov-handoff-3.txt`: the pointed map κ that `KZ2Cohomology.H4` is about, built as the cup square $x \mapsto x \smile x$. Written before the formal proof; each numbered statement is a declaration of `CupSquare` (§4). Proving (H4) for this κ is the next step (§2c) and is not attempted here. Foundations: those of the development (univalence, images and their surjectivity; function extensionality derived). No new axiom; the sphere axiom is not used. Milestone: *a constructed, normalised cup square* κ; that coefficient change along κ is a bijection is the separate theorem (H4). *Status: checked* (64 modules, 0 failed; 2380 declarations; `cup.opens`: nothing else opened). Refereed independently (fresh reader given the note and the library, not the conclusion): constructions correct, interfaces and indices confirmed; corrected here: the centre of S is not definitional (recentring, `contr-at`), lemmas added to (I6), the direction $f = \mathsf{loop}\, c_2$, and the remark on (H4) (the looping of κ is null).

The construction is that of Wärn (*Eilenberg–MacLane spaces and stabilisation in homotopy type theory*, J. Homotopy Relat. Struct. 2023, §4.3): the cup product $K(A,n) \to_* \mathrm{Map}_*(K(B,m), K(C,n+m))$ is obtained by delooping, one level at a time, starting from a bilinear map $A \times B \to C$. We need only $A = B = C = \mathbb{Z}$, multiplication, and the two levels $n = 1, 2$ with $m = 2$; for $A = \mathbb{Z}$ the first level is given by the circle property of $K(\mathbb{Z},1)$ (`CircleProperty`, step §2a), so no coefficient maps $K(a \cdot -, 2)$ and no multiplicativity argument are needed.

1. Conventions and interfaces

As in `doc/pi3-s2-postnikov.pdf`: `isTrunc n X` is "X is $(n-2)$-truncated", `isConn n X` is "X is $(n-2)$-connected"; $K_m = K(\mathbb{Z}, m)$ is `EMMaps.KK[Zab].K m`, pointed by $*$ (`KK.pt m`); $\Omega Y = (y_0 = y_0, \mathsf{refl})$. Paths compose in diagrammatic order, $p \cdot q$ is `trans p q`, by J on q, so $p \cdot \mathsf{refl} \equiv p$. For pointed (X, x_0), (Y, y_0) the pointed mapping type $\mathrm{Map}_*(X,Y) = (F : X \to Y) \times (F\, x_0 = y_0)$ (`Lemma4.PtMap`) is itself pointed, by the constant map $0 = (\lambda_.\, y_0, \mathsf{refl})$.

The construction uses the following, all in the library or proved in §2a:

- (I1) `KK`: $\mathsf{isConn}\,(m+2)\,K_{m+1}$, $\mathsf{isTrunc}\,(m+3)\,K_{m+1}$, and pointed equivalences $e_m : \Omega K_{m+1} \simeq_* K_m$ (`KK.e`).
- (I2) `Lemma4.cor5-map`: if $\mathsf{isConn}\,(c+1)\,X$ and $\mathsf{isTrunc}\,(c+t)\,Y$ then $\mathsf{isTrunc}\,t\,\mathrm{Map}_*(X,Y)$.
- (I3) `DeloopMaps.Maps.Over.cor11` (DW1 Cor. 11): if $\mathsf{isConn}\,(c+2)\,X$ and $\mathsf{isTrunc}\,(2(c+1))\,Y$, then for every pointed $f : \Omega X \to_* \Omega Y$ the type $\mathsf{Unloop}\, f = (F : \mathrm{Map}_*(X,Y)) \times (f = \mathsf{loop}\, F)$ is contractible.
- (I4) §2a, `CircleProperty.Circle`: for pointed (Y, y_0), $\mathsf{ev}_\circ : \mathrm{Map}_*(K_1, Y) \simeq \Omega Y$, $F \mapsto (\mathsf{loop}\, F)\, g$, where g is the generator of ΩK_1 (`pointed-equiv`).
- (I5) `LoopIso.loopPE` (Ω of a pointed equivalence), `PtEquivOps.pt-comp`, `pt-inv`; `LoopCalc.ptcomp`.
- (I6) `FundThm.FT` (fundamental theorem of identity types; the map is transport, so it sends refl to the given element by definition), `FunExtUA.FE.wfe` (a product of contractible types is contractible), the singletons `DeloopMaps.singL` ($(y : Y) \times (y_0 = y)$) and `FunExtUA.sing` ($(w : W) \times (w = w_0)$), `FunExtUA.pathC`, `Fibres.trans-equiv` ($a : x = y$ gives $(y = z) \simeq (x = z)$, $q \mapsto a \cdot q$), `HLevels.trunc-Sigma-contr` (the fibre over the *centre carried by the contractibility proof* decides), `Retracts` (`rt-Sigma`, `rt-trunc`) and `CircleProperty.rt-equiv` (an equivalence as a retract).

The one new general lemma is Lemma 1.

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2. Loops of a pointed mapping type

Lemma 1. *For pointed (X, x_0), (Y, y_0), with $M = \mathrm{Map}_*(X,Y)$ pointed by 0, there is a pointed equivalence*

$$\mathsf{ev} : \Omega M \simeq_* \mathrm{Map}_*(X, \Omega Y), \qquad \mathsf{ev}\,\mathsf{refl} \equiv r = (\lambda_.\,\mathsf{refl}_{y_0}, \mathsf{refl}),$$

the base point of $\mathrm{Map}_*(X,\Omega Y)$, which evaluates a path of pointed maps pointwise: for $P : 0 = G$ and $x : X$, $(\mathsf{ev}\,P)_1\,x = \mathsf{ap}_{\lambda F.\,F_1 x}\,P$ (`ev-ap`, by J on P: both sides are refl *at* refl*).*

Proof. For $G = (k, w) : M$ let

$$R\,G = \big(H : (x : X) \to y_0 = k\,x\big) \times \big(H\,x_0 \cdot w = \mathsf{refl}_{y_0}\big).$$

At $G = 0$ we have $w = \mathsf{refl}$, so $H\,x_0 \cdot w \equiv H\,x_0$ and $R\,0$ is, by definition, $\mathrm{Map}_*(X,\Omega Y)$; let $r = (\lambda_.\mathsf{refl}, \mathsf{refl}) : R\,0$, its base point. By the fundamental theorem of identity types (I6), transport $\mathsf{tr}_G : (0 = G) \to R\,G$, $p \mapsto p_* r$, is an equivalence as soon as the total space $T = (G : M) \times R\,G$ is contractible; take $\mathsf{ev} = \mathsf{tr}_0$. Then $\mathsf{ev}\,\mathsf{refl} \equiv r$, so ev is pointed by refl, and it is an equivalence.

T is contractible. Reordering components (a retract whose round trip is the identity by definition),

$$T \lhd (s : S) \times W\,s, \quad S = (k : X \to Y) \times ((x : X) \to y_0 = k\,x), \quad W(k,H) = (w : k\,x_0 = y_0) \times (H\,x_0 \cdot w = \mathsf{refl}).$$

S is a retract (again by definition) of $(x : X) \to (y : Y) \times (y_0 = y)$, which is contractible by `wfe` and `singL`. The centre produced this way goes through univalence and is not $(\lambda_.\,y_0, \lambda_.\,\mathsf{refl})$ by definition, so we *recentre*: a contractible type is contractible with any given centre a, with contraction $z \mapsto \mathsf{pathC}\,a\,z$ (`contr-at`). With centre $s_0 = (\lambda_.\,y_0, \lambda_.\,\mathsf{refl})$ the fibre is, by definition, $W\,s_0 = W_0 = (w : y_0 = y_0) \times (\mathsf{refl} \cdot w = \mathsf{refl})$. For each w, composition on the left with $a_w = (\mathsf{trans\text{-}refl\text{-}l}\,w)^{-1} : w = \mathsf{refl} \cdot w$ is an equivalence $(\mathsf{refl} \cdot w = \mathsf{refl}) \simeq (w = \mathsf{refl})$ (`trans-equiv`), hence a retract (`rt-equiv`; its round trip is a path, not a definition); by `rt-Sigma`, W_0 is a retract of the singleton $(w : y_0 = y_0) \times (w = \mathsf{refl})$ (`sing`), contractible. Hence $(s : S) \times W\,s$ is contractible (`trunc-Sigma-contr`), and so is T (`rt-trunc`). □

Remark 2. Only the pointed inverse $\mathsf{ev}^{-1} : \mathrm{Map}_*(X,\Omega Y) \to_* \Omega M$ (`pt-inv`) is used in the construction; the equivalence itself (and the formula $\mathsf{ev}\,P = P_* r$) is what (H4) will use to compute loops of $\mathrm{Map}_*(K_2, K_4)$.

3. The construction

Fix $X = K_2$, pointed by $x_0 = *$, and write $M(Y) = \mathrm{Map}_*(K_2, Y)$, pointed by 0.

Construction 3 (the fundamental class)**.** $\iota : K_2 \to_* \Omega^2 K_4$ is the inverse (`pt-inv`) of the pointed equivalence

$$\Omega^2 K_4 \xrightarrow{\Omega e_3} \Omega K_3 \xrightarrow{e_2} K_2$$

(`loopPE`, `pt-comp`).

Construction 4 (level one)**.** Let $\ell = \mathsf{ev}^{-1}\,\iota : \Omega M(\Omega K_4)$ (Lemma 1 for $Y = \Omega K_4$), and let

$$c_1 = \mathsf{ev}_\circ^{-1}\,\ell \;:\; K_1 \to_* M(\Omega K_4)$$

by the circle property (I4) for $Y = M(\Omega K_4)$, $y_0 = 0$. By the section equation, $(\mathsf{loop}\,c_1)\,g = \ell$.

Construction 5 (level two: delooping)**.** Let $f : \Omega K_2 \to_* \Omega M(K_4)$ be the composite of pointed maps

$$\Omega K_2 \xrightarrow{e_1} K_1 \xrightarrow{c_1} M(\Omega K_4) \xrightarrow{\mathsf{ev}^{-1}} \Omega M(K_4)$$

(Lemma 1 for $Y = K_4$). Apply (I3) with $c = 1$: K_2 is 1-connected, $\mathsf{isConn}\,3\,K_2$ (I1), and $M(K_4)$ is 2-truncated, $\mathsf{isTrunc}\,4\,M(K_4)$, by (I2) with $c = 2$, $t = 4$, since $\mathsf{isTrunc}\,6\,K_4$ (I1). So $\mathsf{Unloop}\,f$ is contractible; let

$$c_2 : K_2 \to_* M(K_4), \qquad f = \mathsf{loop}\,c_2$$

be its centre (both components; the direction is that of `Unloop`). Informally, c_2 is $x \mapsto (x \smile -)$; nothing below uses this gloss.

Definition 6 (the cup square)**.** $\kappa : K_2 \to_* K_4$, $\kappa\,x = (c_2\,x)_1\,x$, pointed by the pointing $(c_2\,x_0)_2 : (c_2\,x_0)_1\,x_0 = *$ of the map $c_2\,x_0$.

Theorem 7 (normalisation of c_2)**.** *Let $\bar{c}_1 : K_1 \to_* M(\Omega K_4)$ be c_2 looped and read through the equivalences: $\bar{c}_1 = e_1^{-1}; \mathsf{loop}\,c_2; \mathsf{ev}$ (`pt-inv`, `ptcomp`, Lemma 1 for $Y = K_4$). Then $\mathsf{ev}\,(\mathsf{ev}_\circ\,\bar{c}_1) = \iota$ (Lemma 1 for $Y = \Omega K_4$): looping c_2, applying the mapping-loop equivalences and evaluating on the circle generator g gives back the fundamental class. This records which cup product is constructed.*

Proof. The underlying maps are homotopic: for $k : K_1$, using $f = \mathsf{loop}\,c_2$, $\mathsf{ev} \circ \mathsf{ev}^{-1} = \mathrm{id}$ and $e_1 \circ e_1^{-1} = \mathrm{id}$ (section equations), $(\bar{c}_1)_1\,k = \mathsf{ev}((\mathsf{loop}\,c_2)_1(e_1^{-1}k)) = \mathsf{ev}(f_1(e_1^{-1}k)) = \mathsf{ev}(\mathsf{ev}^{-1}(c_1(e_1(e_1^{-1}k)))) = c_1\,k$. We do not claim a pointed equality: its coherence would be an equation between two paths in $M(\Omega K_4)$, which is a 1-type, not a set. Instead, for pointed maps with an unpointed homotopy H, the loopings are conjugate: $\mathsf{loop}\,\bar{c}_1\,q = c^{-1} \cdot \mathsf{loop}\,c_1\,q \cdot c$ for a loop c of $M(\Omega K_4)$ at 0 (`LoopConj.LoopHomotopy.loop-homotopy`). The type $M(\Omega K_4)$ is pointed equivalent to the loop space $\Omega M(K_4)$ (Lemma 1 for $Y = K_4$), so its loops commute as paths (Eckmann–Hilton, `EH.Abelian.comm`) and conjugation is the identity. Hence $\mathsf{ev}_\circ\,\bar{c}_1 = (\mathsf{loop}\,\bar{c}_1)\,g = (\mathsf{loop}\,c_1)\,g = \ell$ (`evPt-loop`), and $\mathsf{ev}\,\ell = \iota$ by the section equation. □

Remark 8 (canonicity and relation to Wärn). Every choice above is forced by ι and the generator $g = \mathsf{unband}\,1$ of ΩK_1 used in (I4): ev and $\mathsf{ev}_\circ$ are equivalences and $\mathsf{Unloop}\,f$ is contractible, so c_2 is the unique pointed map whose looping is f, and κ depends only on the pointed equivalences e_m of (I1) and on g. In Wärn's construction for the bilinear map $\mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$, level zero is $a \mapsto K(a \cdot -, 2)$, a homomorphism $\mathbb{Z} \to \Omega M(\Omega K_4)$ (after e); for the group $\mathbb{Z}$ a homomorphism is given by the image of 1, $K(1 \cdot -, 2) = \mathrm{id}$, which corresponds to ℓ; the circle property replaces the delooping $K(\mathbb{Z},0) \to K(\mathbb{Z},1)$ (DW1 Cor. 12 and its multiplicativity condition). Level two is Wärn's delooping step, by (I3). The pointing of κ could also be taken from the pointing $c_2\,x_0 = 0$ of c_2, evaluated at x_0; the two pointings differ by a path, which (H4) may have to relate. We fix the pointing of the map $c_2\,x_0$, a component, which needs no lemma.

Remark 9 (what (H4) will need; the looping of κ is null). The characterising equations of the construction are $\mathsf{ev}\,\mathsf{refl} \equiv r$, $(\mathsf{loop}\,c_1)\,g = \ell$ (section equation, then `evPt-loop`) and $f = \mathsf{loop}\,c_2$ (I3). They describe c_2, not κ directly: the looping of κ is $\mathsf{loop}\,\kappa\,p = (\mathsf{ap}_{c_2}p)(x_0) \cdot \mathsf{ap}_{c_2 x_0}p$, whose first factor is trivial because the points of $M(K_4)$ are pointed maps and $c_2\,x_0 = 0$. Indeed $\Omega\kappa : K_1 \to_* K_3$ is null ($[K(\mathbb{Z},1), K(\mathbb{Z},3)]_* = H^3(S^1) = 0$). So looping cannot detect the class of κ. (H4) must identify $[\kappa]$ in $[K_2, K_4]_*$ (the set of deloopings of the zero map $K_1 \to_* K_3$, `doc/pi3-s2-postnikov.pdf`) using c_2 as a map $K_2 \to_* M(K_4)$, i.e. its effect on π_2 (by the equations above, through c_1 and ι), together with the diagonal; and it must prove naturality of $g \mapsto [\kappa; K(g,4)]$ in the coefficients, which this construction does not provide.

4. Correspondence with the formal proof

File `CupSquare.seg` (after `CircleProperty` in `All.seg`). File `sag/pi3-s2-postnikov/CupSquare.seg`; `-opens` specification `cup.opens`.

- Lemma 1: `contr-at`; module `PtMapLoops[X,x0,Y,y0]`: `pt0`, `R`, `r`, `R-pt0` (by `refl`), `S`, `s0`, `W`, `S-rt`, `S-contr`, `W0-rt`, `T-rt`, `T-contr`, `ev` (`FT.tr pt0`), `ev-ap`, `ev-pt` (pointed equivalence), `unev` (its pointed inverse), `unev-un`, `ev-unev` (then `unev` is locked).

- Construction 3: `iota-eq`, `iota`.
- Construction 4: `ell`, `ell-ev`, `c1`, `c1-ev` (then `c1` is locked).
- Construction 5: `e1` (locked), `f`, `tM4`, `cell`, `c2`, `c2-loop` (then `c2` is locked).
- Definition 6: `kappa : PtMap KZ2 kz2 KZ4 kz4`.
- Theorem 7: `e1inv`, `e1-back`, `c1bar`, `c1bar-hom`, `AB` (`EH.Abelian`), `conj-triv`, `LH` (`LoopConj.LoopHomotopy`
`cup-norm`.